

Serie 28

BILINEAR FORMS, JORDAN NORMAL FORM

1. Consider the real matrix

$$A = \begin{pmatrix} 14 & -13 & 8 \\ -13 & 14 & 8 \\ 8 & 8 & -7 \end{pmatrix} \in M_{3 \times 3}(\mathbb{R}).$$

Find a matrix $P \in O_3(\mathbb{R})$ such that $P^{-1}AP$ is diagonal.

2. Let V be an n -dimensional vector space over a field K . Show the following statements using the lemma about generalized eigenspaces seen in the lectures, but without using the Jordan Normal Form theorem:

- (a) Suppose that $N \in \text{End}(V)$ is nilpotent. Then 0 is an eigenvalue of N and it is the only one.
- (b) If $N \in \text{End}(V)$ is nilpotent, then $N^n = O_{n \times n}$. In other words, the nilpotency index of N is smaller or equal to $\dim(V)$.
- (c) Suppose that $N \in \text{End}(V)$ is nilpotent and assume that $p_N(x)$ splits as a product of linear factors in $K[x]$. Then $p_N(x) = (-x)^n$.
- (d) Let $T \in \text{End}(V)$ and assume that $p_T(x)$ splits into linear factors in $K[x]$. Let $\eta \in K$ and define $S = T - \eta \text{Id}_V$. Then $p_S(x)$ also splits into a product of linear factors over $K[x]$. In fact, $p_S(x) = p_T(x + \eta)$.
- (e) Let $T \in \text{End}(V)$ and assume that $\lambda \in K$ is the only eigenvalue of T and that $p_T(x)$ splits as a product of linear factors in $K[x]$. Define $N = T - \lambda \text{Id}_V$. Then $p_N(x) = (-x)^n$, and $N^n = O_{n \times n}$.

3. Determine the Jordan normal form of the following matrix over $\mathbb{R}$ and over $\mathbb{F}_3$:

$$A := \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

4. Determine the Jordan normal form and the corresponding base change matrices of the real matrix

$$A := \begin{pmatrix} 2 & 2 & 2 & 2 \\ 0 & 3 & 0 & 2 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Single Choice. In each exercise, exactly one answer is correct.

1. Let f be an endomorphism of a finite-dimensional vector space V and let λ be an eigenvalue of f . Which statement is generally false?
 - (a) Every eigenvector of f corresponding to the eigenvalue λ lies in the eigenspace $\tilde{\text{Eig}}_f(\lambda)$.
 - (b) Every vector in $\tilde{\text{Eig}}_f(\lambda)$ is an eigenvector of f corresponding to the eigenvalue λ .
 - (c) The generalized eigenspace $\tilde{\text{Eig}}_f(\lambda)$ is not the zero space.
 - (d) For every eigenvalue μ of f with $\mu \neq \lambda$, we have $\tilde{\text{Eig}}_f(\mu) \cap \tilde{\text{Eig}}_f(\lambda) = \langle 0 \rangle$.
2. For every endomorphism f of an n -dimensional vector space V whose characteristic polynomial factors into linear factors, and every eigenvalue λ of f we have:
 - (a) $\tilde{\text{Eig}}_f(\lambda) = \text{Kern}(f - \lambda \text{id}_V)$.
 - (b) $\dim(\tilde{\text{Eig}}_f(\lambda)) = 1$.
 - (c) $\dim(\tilde{\text{Eig}}_f(\lambda)) = n$.
 - (d) $\tilde{\text{Eig}}_f(\lambda) = \text{Kern}((f - \lambda \text{id}_V)^n)$.
3. The generalized eigenspace of the real matrix $A := \begin{pmatrix} 2 & 3 & -1 & 5 \\ 0 & 2 & -3 & 1 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 3 \end{pmatrix}$ with respect to $X - 2$ is
 - (a) One-dimensional
 - (b) Two-dimensional
 - (c) Three-dimensional
 - (d) Four-dimensional
4. Let A be a 3×3 matrix with $A \neq 0$ and $A^2 = 0$. Then, the Jordan normal form of A has
 - (a) 1 Jordan block.
 - (b) 2 Jordan blocks.
 - (c) 3 Jordan blocks.
 - (d) It depends on the exact matrix A .

Multiple Choice Fragen

1. Which of the following statements is **true**: For arbitrary integers $n > m \geq 1$, there exists a square matrix with...
 - (a) characteristic polynomial $X^m + X^n$.
 - (b) minimal polynomial X^m and characteristic polynomial X^n .
 - (c) minimal polynomial $X^m \cdot (X^n - 1)$.